## Week 6-8

**Focus**: Explore different text encoding method with a linear probe as downstream and observe the difference in performance.
**Progress targets**:

1. **TF-IDF**
    - Use TF-IDF (w/wo n-grams (n=2,3,5)) to encode the abstract and use a logistic regression as the prediction head.
    - You can try to use explore some traditional NLP techniques like stopwords, lemming, etc. to do the preprocessing.
    - Please record the result and try to explain what makes the difference.
2. **Word2vec**
    - Use a pretrained word2vec model (e.g., word2vec-google-news-300) to encode the abstract and a logistic regression as the prediction head.
    - Record the result.
    - (optional) if you are interested you can try your own model on the training set and do the same experiment and see the difference.
3. **Modern encoding method**
    - Try modern embedding model (i.e., EmbeddingGemma, GPT4) to encode and do the same experiment.

**By end of Week 8, you should have:**

Finish all the experiment and present the performance under the 3 different settings.

## Week 8-12

Focus: Try some more modern methods.
**Progress targets:**

1. **Frozen transformer**
    - Load **DistilBERT** (or BERT-base if VRAM allows), **freeze encoder**, train **linear head** only.
2. Light fine-tuning
    - Unfreeze the last few layers or full model if VRAM is ok and fine-tune the model.
    - Try LORA if the fine-tuning is not stable.
3. Record all the results and summarize what we have done this semester.

**By end of Week 12, you should have:**

Finish all the experiment and have a detailed report of all techniques we tried this term.